

## Lab 3 – Create your Virtual Private Cloud (VPC)

Student Full Name Greemesh Basnet \_\_\_\_\_/10 Marks

### Objective

To understand how a Virtual Private Cloud can be configured.

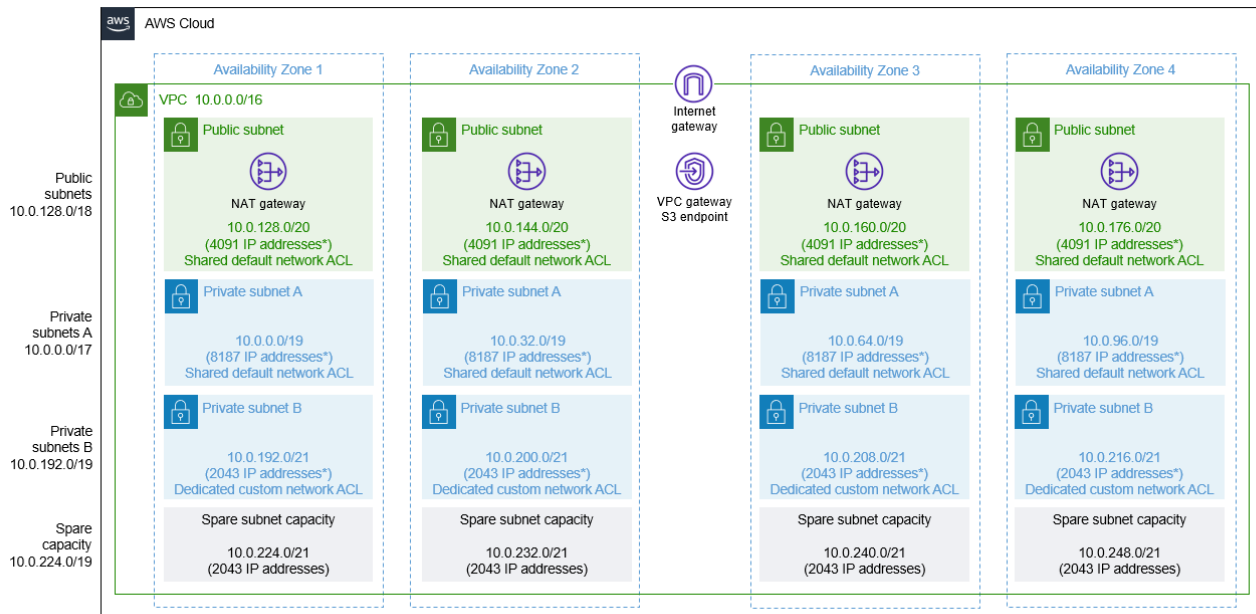
### Background

- Use the material posted from week 3 and 4 to verify the Cloud and Network Functions Virtualization concepts.
- In CSN205 you have learned about IP addressing. Use this knowledge to solve and verify the addresses of your own VPC architecture.
- In this lab guide, you need to use your own AWS account. **IMPORTANT: Stop all your instances each time your activities end and restart them again next time. Avoid charges to your AWS account.**
- When you create any entity such as security group, IGW, route table, etc. put a meaningful description/name.

This Architecture shown below uses a virtual networking environment in the AWS Cloud. You can use it and the link below as a reference for this project:

[Amazon Virtual Private Cloud on AWS](#)

You are not building the exact environment show below; it is an example similar to the network you will create in this lab session.



\*Five IP addresses in each subnet CIDR block are reserved and unavailable for use.

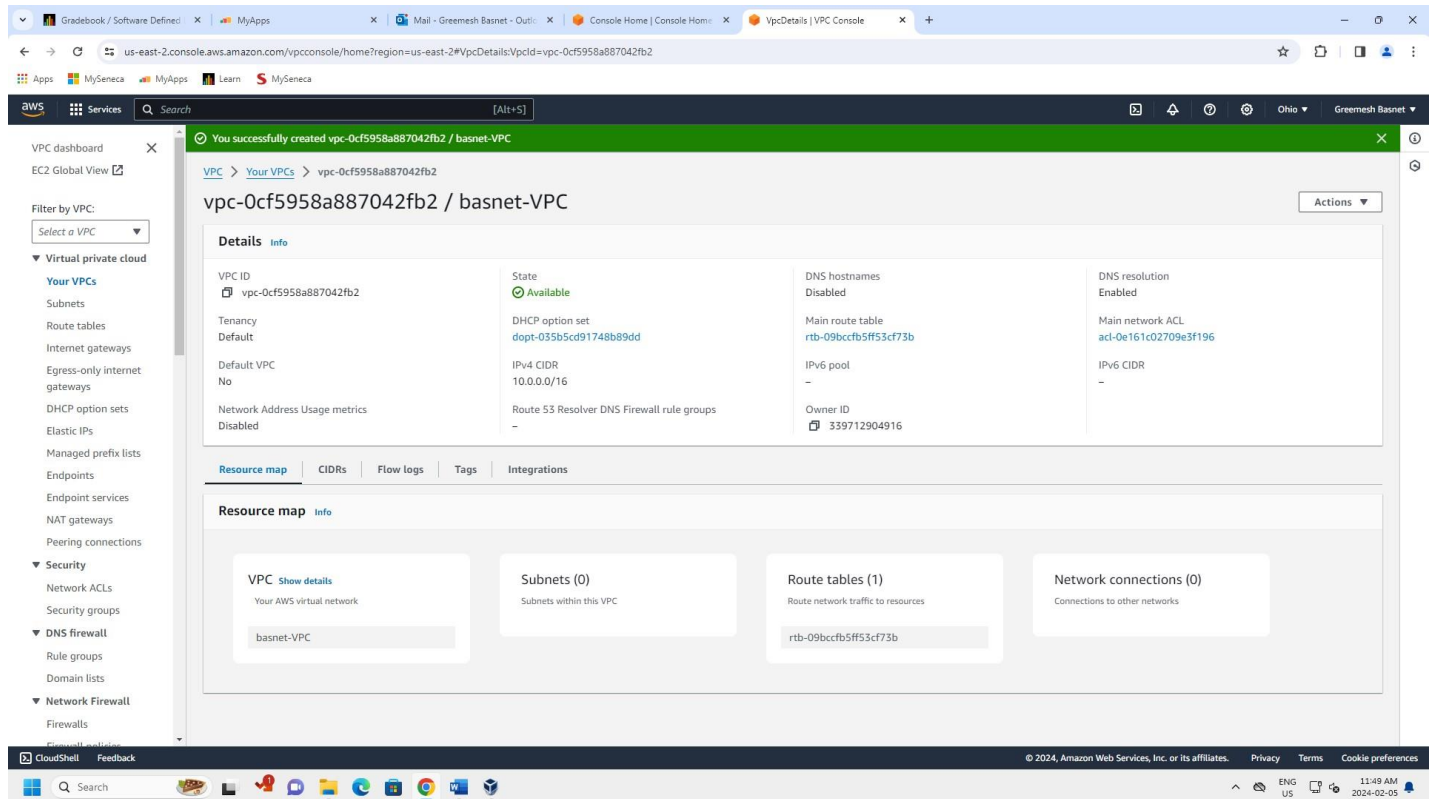
**Instructions**

1. Instructions will be given in class about a AWS account.
2. **Use a different font and color to write your answers.**
3. Do not erase anything from this document.
4. Resize the screenshots to make them easily readable.
5. Make your name visible on each one of your inserted screenshots, as well as the date and time of your host machine. Screenshots without these two conditions are not accepted.

**Procedure**

Create a VPC with IPv4 IP block of 10.0.0.0/16 and name it: `<my_last_name>-VPC` **Take the screen shot and insert it here.** (1 mark)

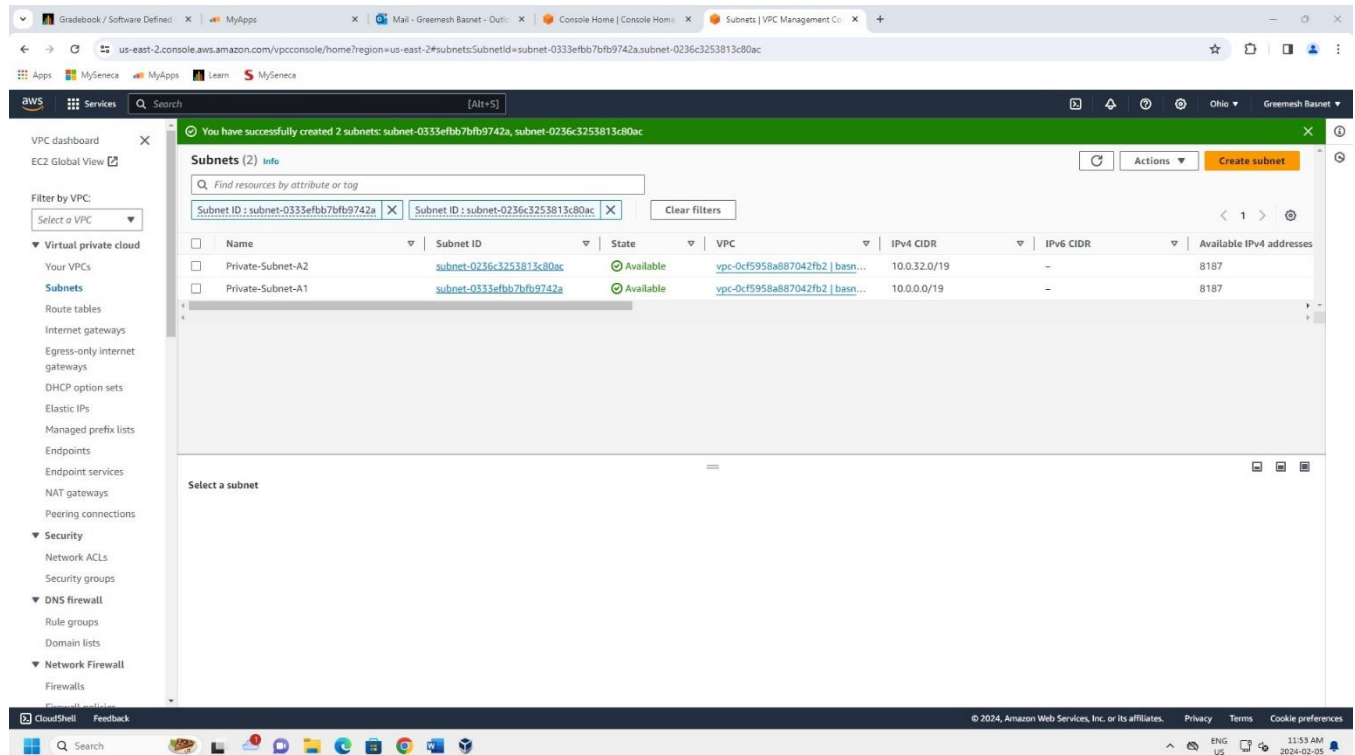
**Stop** all your instances each time your activities end and restart them again



- 1) Create two private subnets: **Take the screen shot and insert it here.** (2 marks)

Private-Subnet-A1: 10.0.0.0/19

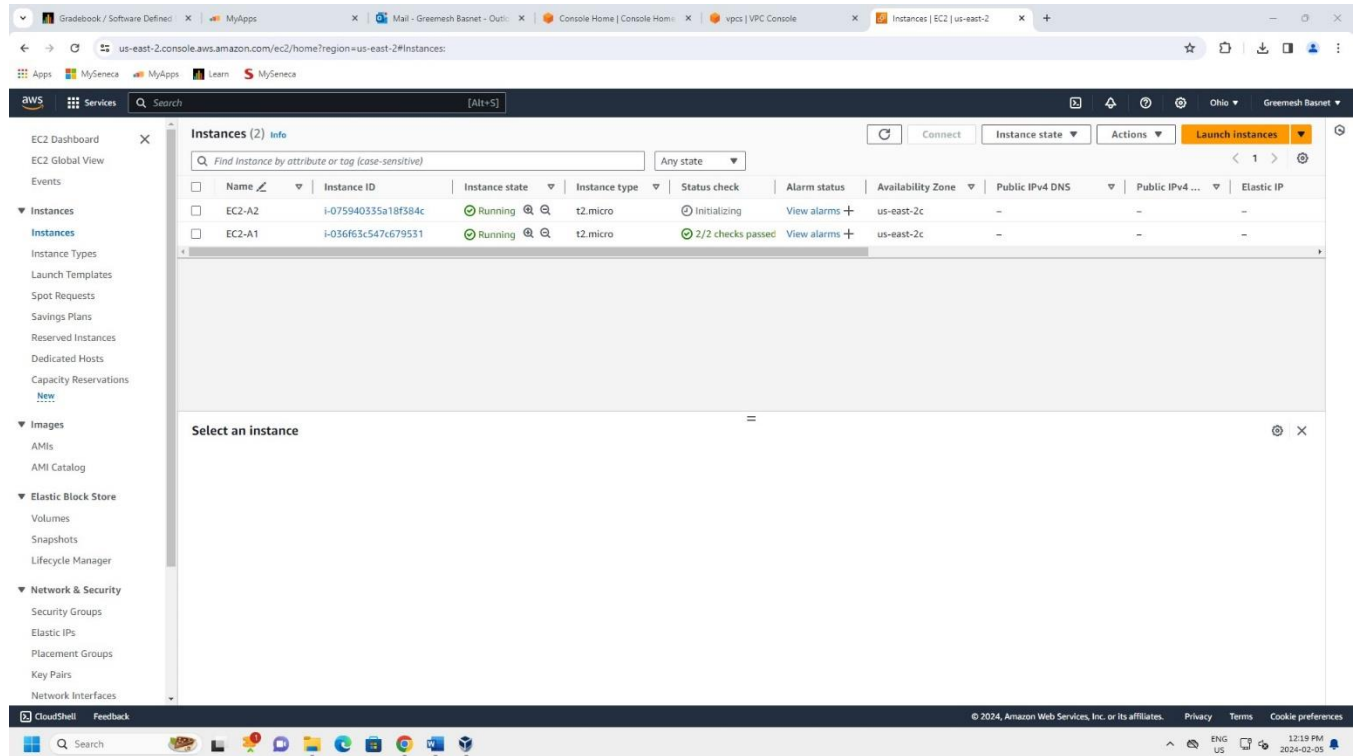
Private-Subnet-A2: 10.0.32.0/19



- 2) Create one ubuntu EC2 instance, free tier in each of these subnets. **Take the screen shot and insert it here.** (2 marks)

EC2-A1 in Private-Subnet-A1

EC2-A2 in Private-Subnet-A2

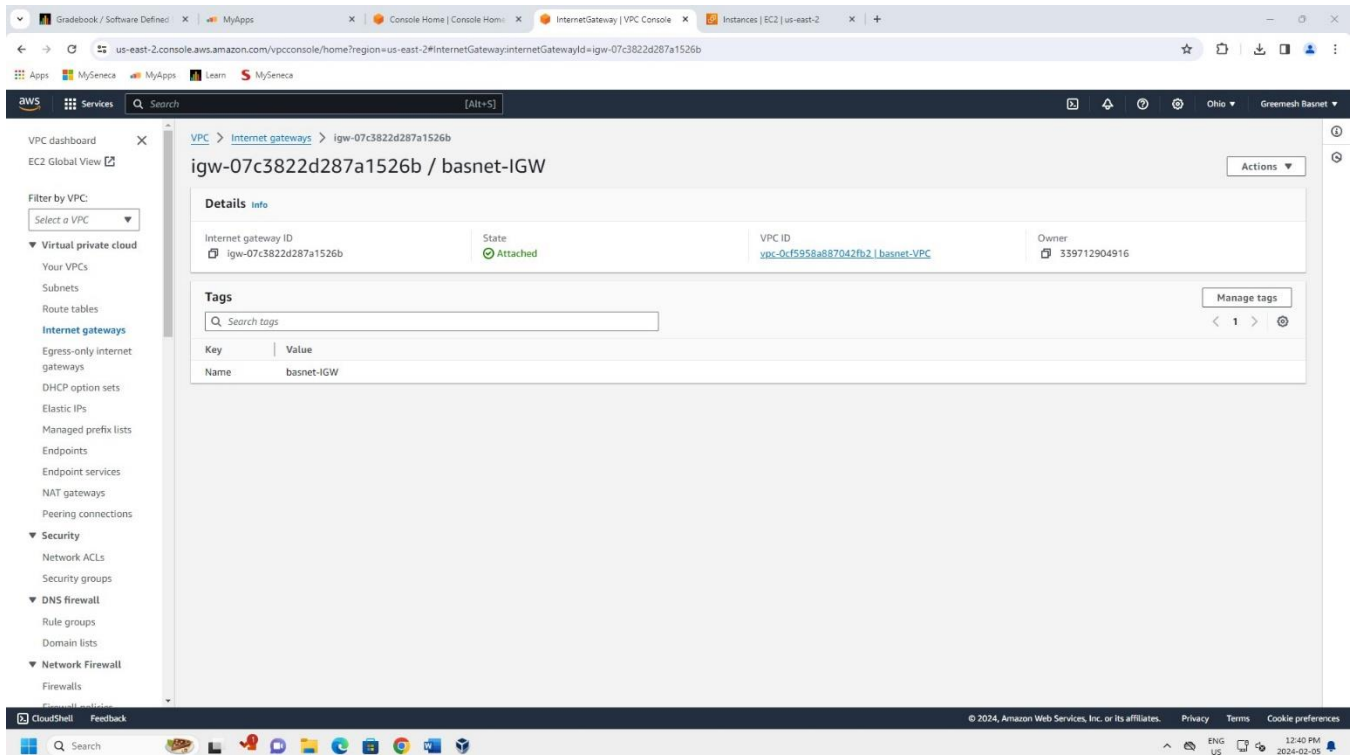


- 3) Can you find any “public-dns-name” in their description?  
If not, how you can obtain it?

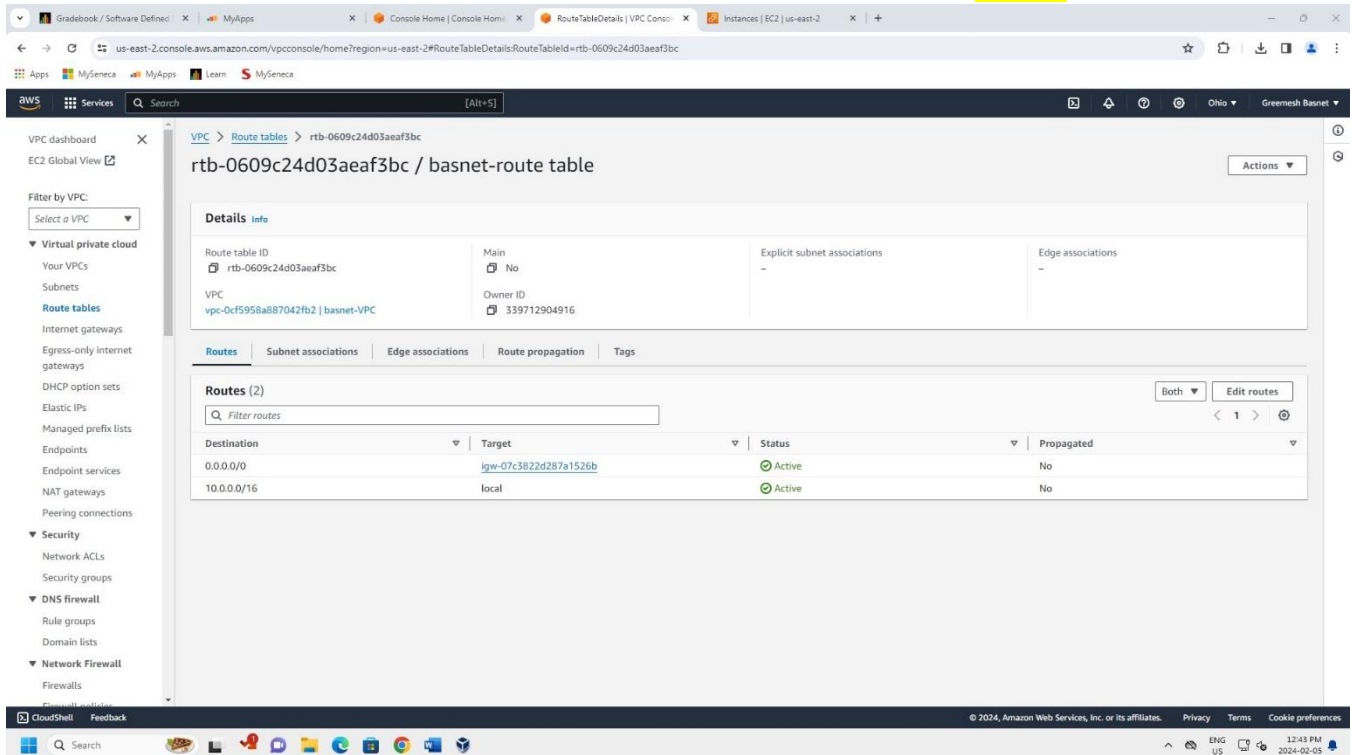
*Note: You need “public-dns-name” to connect remotely to those instances.*

*Answer: I can obtain public-dns-name by giving Elastic IP to the instances.*

4. Create an IGW in your VPC and attach it. **Take the screen shot and insert it here. (1 mark)**



- 4) Create route table to allow 0.0.0.0/0 to internet. **Take the screen shot and insert it here. (1 mark)**



- 5) In security groups of EC2 instances configure SSH connection TCP port 22 to be allowed in inbound and outbound.

**Take the screen shot and insert it here. (1 mark)**

The image displays two screenshots of the AWS Management Console, specifically the 'Security Groups' page for a security group named 'sg-0115532daaa664e42 - basnet-securitygroup'.

**Top Screenshot:** Shows the 'Outbound rules' tab. The 'Outbound rules (1)' table contains one rule:

| Name | Security group rule... | IP version | Type | Protocol | Port range | Destination | Description |
|------|------------------------|------------|------|----------|------------|-------------|-------------|
| -    | sgr-0d787dd6b339a9...  | IPv4       | SSH  | TCP      | 22         | 0.0.0.0/0   | -           |

**Bottom Screenshot:** Shows the 'Inbound rules' tab. The 'Inbound rules (1)' table contains one rule:

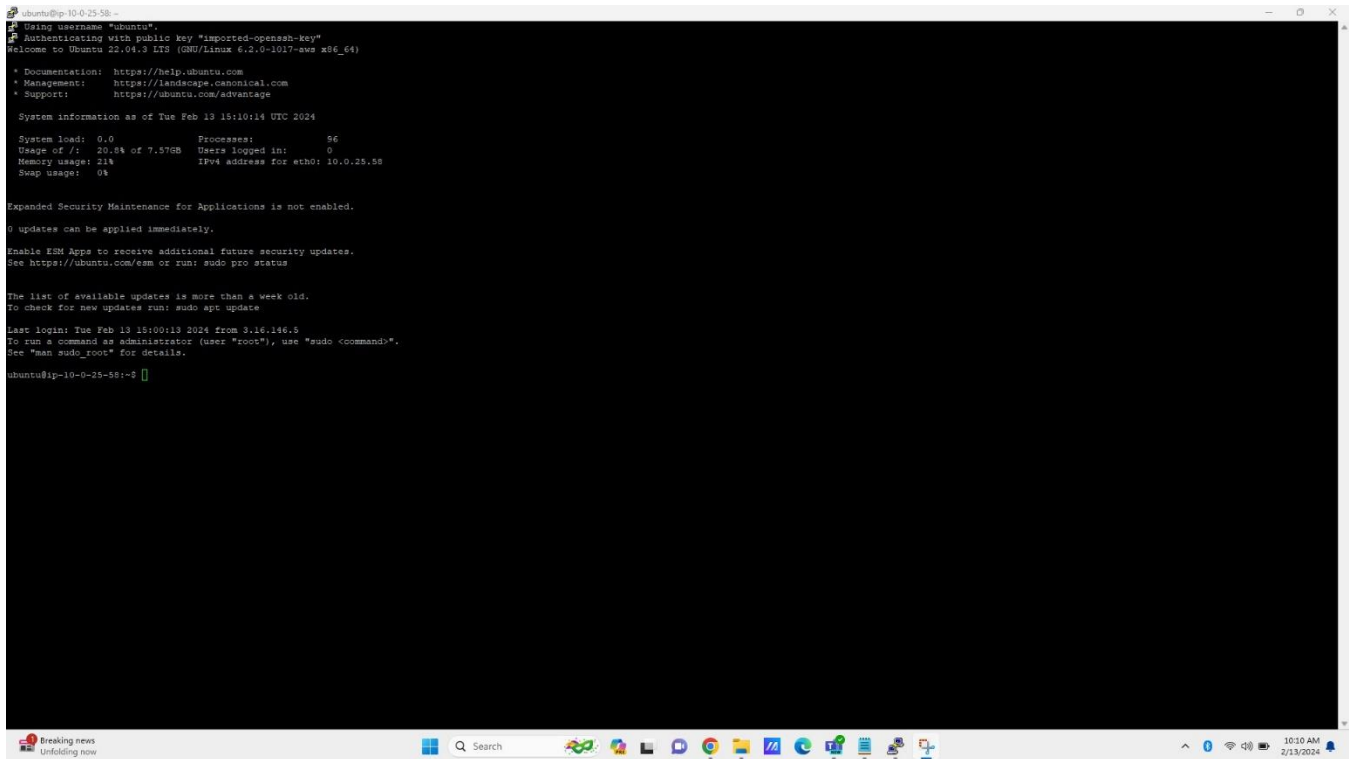
| Name | Security group rule... | IP version | Type | Protocol | Port range | Source    | Description |
|------|------------------------|------------|------|----------|------------|-----------|-------------|
| -    | sgr-0f6b30bdf2eed513f  | IPv4       | SSH  | TCP      | 22         | 0.0.0.0/0 | -           |

Both screenshots show the 'Details' section with the following information:

- Security group name: basnet-securitygroup
- Security group ID: sg-0115532daaa664e42
- Description: ssh
- VPC ID: vpc-0cf5958a887042fb2
- Owner: 339712904916
- Inbound rules count: 1 Permission entry
- Outbound rules count: 1 Permission entry

- 6) Connect to EC2-A1 from your computer by Putty. **Take the screen shot and insert it here.** (2 marks)

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/putty.html>



The screenshot shows a terminal window titled 'ubuntu@ip-10-0-25-58'. The terminal output includes the following text:

```
Using username "ubuntu".
Authenticating with public key "imported-openssh-key"
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 6.2.0-1017-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Tue Feb 13 15:10:14 UTC 2024

System load:  0.0          Processes:    96
Usage of /:   20.8% of 7.57GB    Users logged in:  0
Memory usage: 21%          IPv4 address for eth0: 10.0.29.58
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Tue Feb 13 15:00:13 2024 from 3.16.146.5
To run a command as administrator (user "root"), use "sudo ".
See "man sudo_root" for details.

ubuntu@ip-10-0-25-58:~$
```

The terminal window is displayed within a desktop environment. The taskbar at the bottom shows various application icons, including a search bar, file explorer, and web browser. The system clock in the bottom right corner indicates the time is 10:10 AM on 2/13/2024.

- 7) Username is Ubuntu

8) Ping EC2-A2 from EC2-A1. Take the screen shot and insert it here. (5 marks)

```
ubuntu@ip-10-0-25-58:~$ sudo apt update
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 6.2.0-1017-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

System information as of Tue Feb 13 15:10:14 UTC 2024

System load:  0.0               Processes:    96
Usage of /:   20.8% of 7.57GB   Users logged in:  0
Memory usage: 21%              IPv4 address for eth0: 10.0.25.58
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo apt install esm-apps

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Tue Feb 13 15:00:13 2024 from 3.16.166.8
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-10-0-25-58:~$ ping 10.0.34.91
PING 10.0.34.91 (10.0.34.91) 56(84) bytes of data:
64 bytes from 10.0.34.91: icmp_seq=177 ttl=64 time=0.507 ms
64 bytes from 10.0.34.91: icmp_seq=178 ttl=64 time=0.517 ms
64 bytes from 10.0.34.91: icmp_seq=179 ttl=64 time=0.583 ms
64 bytes from 10.0.34.91: icmp_seq=180 ttl=64 time=0.587 ms
64 bytes from 10.0.34.91: icmp_seq=181 ttl=64 time=0.477 ms
64 bytes from 10.0.34.91: icmp_seq=182 ttl=64 time=0.527 ms
64 bytes from 10.0.34.91: icmp_seq=183 ttl=64 time=0.600 ms
64 bytes from 10.0.34.91: icmp_seq=184 ttl=64 time=0.526 ms
64 bytes from 10.0.34.91: icmp_seq=185 ttl=64 time=0.465 ms
64 bytes from 10.0.34.91: icmp_seq=186 ttl=64 time=0.472 ms
64 bytes from 10.0.34.91: icmp_seq=187 ttl=64 time=0.407 ms
64 bytes from 10.0.34.91: icmp_seq=188 ttl=64 time=0.506 ms
64 bytes from 10.0.34.91: icmp_seq=189 ttl=64 time=0.461 ms
64 bytes from 10.0.34.91: icmp_seq=190 ttl=64 time=0.511 ms
64 bytes from 10.0.34.91: icmp_seq=191 ttl=64 time=0.545 ms
64 bytes from 10.0.34.91: icmp_seq=192 ttl=64 time=0.457 ms
64 bytes from 10.0.34.91: icmp_seq=193 ttl=64 time=0.461 ms
64 bytes from 10.0.34.91: icmp_seq=194 ttl=64 time=0.450 ms
64 bytes from 10.0.34.91: icmp_seq=195 ttl=64 time=0.496 ms
^C
--- 10.0.34.91 ping statistics ---
195 packets transmitted, 19 received, 0.02564% packet loss, time 190654ms
rtt min/avg/max/mdev = 0.407/0.503/0.600/0.049 ms
ubuntu@ip-10-0-25-58:~$
```

#### To submit:

- Be sure that you wrote your full name at the beginning of this document. (10% deducted if you do not write your name)
- Rename this file as **LastNameFirstName\_CSN305Lab3** and save it as PDF. Upload this document including all your answers and screenshots. (10% deducted if this file is missed in your submission or wrong name is used)